الجامعة السورية الخاصة
SYRIAN PRIVATE UNIVERSITY

كلية هندسة الذكاء الاصطناعي
FACULTY OF ARTIFICIAL INTELLIGENCE ENGINEERING

Design and Implementation of an Autonomous Vision-Guided Robotic Manipulation System for Intelligent Jigsaw Puzzle Assembly Using a 4-DOF Manipulator

Prepared by:

Hazem Marwan Hamze

Farah Mostafa Al-Mokdad

Supervised by:

Dr. Samer Hussam Eddin

Eng. Alhussain Tofailea

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الملخص

يهدف هذا المشروع إلى تصميم وتنفيذ نظام روبوتي مستقل موجه بالرؤية الحاسوبية للتجميع الذكي لقطع أحجية الصور الجدارية (البازل) باستخدام ذراع روبوتية مفصلية ذات أربع درجات حرية (4-DOF)، يسعى العمل إلى معالجة تحديات المعالجة الآلية للأجسام المسطحة والدقيقة عبر تقديم حل مدمج يربط بين خوارزميات الرؤية الحاسوبية ونظم التحكم الحركي.

تضمن الجانب الهندسي للمشروع النمذجة الكينماتيكية وتحديد الأبعاد الهيكلية للذراع للوصول إلى مساحة عمل خالية من نقاط الانفراد، مع اعتماد منفذ نهائي يعتمد الشفط الهوائي (Vacuum Suction) لتفادي إزاحة القطع المجاورة أثناء الالتقاط.

لتحقيق رؤية شاقوليه خالية من الظلال وتأثيرات الإضاءة المتغيرة، تم تزويد الذراع بكاميرا مدمجة على المنفذ النهائي بوضعية (Eye-in-Hand).

تعتمد طبقة الرؤية الحاسوبية المطورة ضمن بيئة ROS 2 Humble ومكتبة OpenCV على سلسلة معالجة متكاملة تبدأ بتحويل فضاء الألوان من RGB إلى HSV للحد من الحساسية لتغيرات الإضاءة، يليها تطبيق العزل اللوني الثنائي(Binary Masking) واستخراج الحواف(Find Contours)، وصولاً إلى حساب المركز الهندسي بدقة البيكسل وفي الوقت الحقيقي (Real-Time).

أظهرت نتائج المحاكاة الفيزيائية ضمن بيئة Gazebo استقراراً ديناميكياً عالياً للذراع ودقة متناهية في استخراج إحداثيات الأهداف لحظياً، مما أثبت نجاح مرحلة إثبات المفهوم.

يؤسس هذا الإنجاز للمرحلة القادمة المتمثلة في تطبيق الكينماتيكا العكسية(Inverse Kinematics) والتخطيط الحركي باستخدام إطار MoveIt 2 لتنفيذ عمليات الالتقاط والوضع وإعادة التجميع الكامل بالواقع.

Abstract

This project aims to design and implement an autonomous vision-guided robotic manipulation system for intelligent jigsaw puzzle assembly using a 4-degree-of-freedom (4-DOF) articulated manipulator. The work addresses the challenges of automated handling of flat, tightly grouped objects by developing an integrated framework that bridges computer vision algorithms with robotic motion control.

The engineering development involves kinematic modeling and spatial link dimensioning to achieve a singularity-free workspace. A micro vacuum suction cup is selected as the end-effector to enable precise top-down grasping without displacing adjacent puzzle pieces. To ensure a clear, shadow-free vertical line of sight, an Eye-in-Hand camera configuration is integrated directly onto the end-effector.

The vision processing pipeline, implemented within the ROS 2 Humble framework using OpenCV, transforms raw image frames from the BGR to the HSV color space to minimize sensitivity to ambient lighting variations. This is followed by binary color segmentation, contour detection, and real-time target extraction centroid pixel-level precision.

Simulation results obtained in the Gazebo physical simulator demonstrate high dynamic stability of the manipulator and precise real-time target feature extraction, successfully establishing a robust Proof of Concept (PoC). This work lays a solid foundation for the upcoming implementation phases, which will involve applying Inverse Kinematics (IK) and trajectory planning via MoveIt 2 to execute autonomous pick-and-place assembly operations.

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Glossary	
Degrees of Freedom (DOF)	درجات الحرية
Robotic Arm	ذراع روبوتية
Computer Vision	الرؤية الحاسوبية
Eye-in-Hand Camera	كاميرا مثبتة على اليد / الروبوت
OpenCV (Open Source Computer Vision Library)	مكتبة الرؤية الحاسوبية مفتوحة المصدر
ROS 2 (Robot Operating System 2)	نظام تشغيل الروبوتات الإصدار الثاني
Gazebo	بيئة محاكاة فيزيائية ثلاثية الأبعاد
MoveIt 2	حزمة تخطيط المسارات والتحكم الحركي
URDF (Unified Robot Description Format)	تنسيق وصف الروبوت الموحد
Forward Kinematics (FK)	الحركية الأمامية
Inverse Kinematics (IK)	الحركية العكسية
Pick-and-Place	الالتقاط والتوضع (أو النقل)
Jigsaw Puzzle	لعبة بازل (تركيب القطع)
Closed-Loop Control	تحكم مغلق الحلقة
Open-Loop Control	تحكم مفتوح الحلقة
Trajectory Planning	تخطيط المسار الحركي
Collision Avoidance	واجهة المستخدم الرسومية بايثون
CustomTkinter	واجهة المستخدم الرسومية بايثون
Rviz2	أداة التصور والتحويل ثلاثية الأبعاد
Emergency Stop (E-Stop)	زر التوقف الطارئ

Chapter 1: Introduction

1.1 Project Definition:

This project is an integrated intelligent system for handling and assembling puzzle pieces using a 4-Degree-of-Freedom (4-DOF) robotic arm. The system relies on the ROS 2 Humble and Gazebo virtual simulation environments to design and represent the arm's motion, while path planning and collision avoidance are handled via MoveIt 2.
Furthermore, the system incorporates a core computer vision component using the OpenCV library and Python. A camera mounted on the arm processes images in real-time to recognize puzzle pieces, accurately determine their coordinates and orientation angles (Pose Estimation). This enables the robotic arm to pick up the piece, guide it with high precision to its designated location, and automatically complete the puzzle.

2.1 Project Description:

The system consists of three main parts:
1- Simulation & Control System:
It relies on the ROS 2 Humble framework and the Gazebo simulation environment to model the motion of the 4-DOF robotic arm using URDF files. This includes forward and inverse kinematics calculations, as well as utilizing MoveIt 2 for precise motion path planning and collision avoidance.
2- Computer Vision System:
It utilizes an eye-in-hand camera mounted on the robotic arm to process images in real-time via OpenCV and Python. This is done to recognize puzzle pieces, filter out noise using morphological operations, extract the precise coordinates of the piece's center, and publish them via ROS 2 Topics.
3- Execution & Integration Unit:
It bridges the outputs of the vision system (coordinates and orientation angles) with the inputs of the motion planner to guide the robotic manipulator toward the target, performing automated and synchronized pick-and-place and puzzle assembly operations.

3-1 Problem Statement and Its Practical and Academic Significance

The Problem:

Traditional manual selection and assembly of puzzle pieces and small parts suffer from extreme slowness, high vulnerability to repeated human errors, as well as the difficulty of instantaneous visual recognition of piece shapes and precise coordinate determination to guide automated pick-and-drop actions without an intelligent control system linking computer vision with robotic arm motion.

Why It Is Worth the Time and Effort:

This project is worth the time and effort because it provides a practical and academic solution for automating precise handling operations. It offers a comprehensive simulation environment to test and develop visual recognition algorithms and motion planning using a limited degrees-of-freedom robotic arm (4-DOF), thereby reducing costs and risks, and contributing to the development of an intelligent system that keeps pace with the aspirations of modern production line automation.

4-1 The Scientific Problem & Research Gap:

The scientific problem addressed by this project lies in the existing gap between theoretical solutions for jigsaw puzzle assembly and physical robotic control in real-world and simulation environments. Most previous research is either limited to theoretical two-dimensional image processing without physical application, or relies on open-loop control systems and fixed trajectories that lack the ability to compensate for localization errors and illumination deviations during pick-and-place operations.

This research aims to address this problem by integrating a closed-loop Eye-in-Hand vision system with a dynamic motion planning framework (ROS 2 & MoveIt 2) using a 4-DOF robotic arm, which ensures real-time s *correction of the 3D position and orientation (X, Y, Z, θ) of puzzle piece and successfully assembles them within the physical simulation environment (Gazebo).

1-5 Project Objectives:

1-5-1 Main Objectives:

Design an intelligent and integrated automated system for handling and assembling puzzle pieces using a 4-DOF robotic arm.

Leverage modern virtual simulation environments, ROS 2 Humble and Gazebo, to model the robot and test its kinematic behavior.

Achieve high precision in motion planning and virtual obstacle avoidance using MoveIt 2.

Integrate computer vision techniques (OpenCV) with robotic control systems for automated visual object recognition.

1-5-2 Secondary Objectives:

Build and develop a 3D robot model in URDF format with a virtual camera mounted in an Eye-in-Hand configuration.

Process images in real-time to extract contours and determine precise pixel coordinates for the puzzle pieces.

Develop software nodes in ROS 2 (Nodes & Topics) to seamlessly publish and subscribe to coordinate data.

Execute automated pick-and-place and puzzle assembly operations fully without human intervention.

1-6 Project Significance:

Enhancing Precision Automation:
Contributes to automating complex handling and assembly operations with high precision, thereby reducing human errors in repetitive manual tasks.

Keeping Pace with Smart Industry Trends (Industry 4.0): Reflects the importance of integrating robotic systems with visual intelligence systems to execute smart tasks resembling modern production lines.

Effective Integration Between Vision and Control: Highlights the practical value of directly linking image processing algorithms (OpenCV) with robot motion planning schemes (MoveIt 2) to execute tasks based on visual feedback.

Cost Reduction and Prototyping Facilitation: Relies on open-source simulation environments (ROS 2 / Gazebo), enabling low-cost system testing, development, and programming without hardware risks.

Academic and Research Value: Provides an integrated experimental platform for control and robotics engineering students to apply concepts of kinematics, motion planning, and image processing in a realistic environment.

1-7 Project Problems and Challenges:

1. Technical and Engineering Challenges:

Computer Vision Accuracy and Image Processing: Puzzle piece recognition and edge extraction are affected by varying virtual lighting conditions or color overlap in the simulation environment.

Synchronization and Communication Latency: Potential slight delays in transferring coordinates and program data between ROS 2 nodes and the Gazebo simulation system.

Limited Degrees of Freedom (4-DOF): Kinematic constraints resulting from the arm having only 4 degrees of freedom, which poses challenges in achieving proper orientation and reaching complex angles.

Motion Planning Errors: Difficulty in calculating and planning safe trajectories without the arm colliding with surrounding obstacles via MoveIt 2.

2. Operation and Simulation Challenges:

Dynamic Parameters Tuning: The need for precise tuning of friction and mass & inertia properties in URDF files to closely mimic reality.

Spatial Sensing Accuracy: The challenge of converting 2D pixel coordinates into precise 3D spatial coordinates.

Software Node Stability: Avoiding sudden system halts upon exceptions or errors in broadcasting image topics.

3. Safety and Standards Challenges:

Safe Joint Control: The necessity to prevent sudden or uncalculated movements that could exceed physical joint limits.

End-Effector Safety: Ensuring the reliability and stability of gripping and assembling puzzle pieces without slipping during motion.

Open-Source Software Standards: Dealing with potential limitations and vulnerabilities in the ROS 2 packages used.

4. Economic and Practical Challenges:

Computational Cost: The need for high processing resources (a powerful GPU and CPU) to run Gazebo and MoveIt 2 simulations smoothly and without lag.

Future Scalability: The difficulty of upgrading the system later from 4 degrees of freedom to more complex arms without a complete architectural redesign.

Reliability in the Working Environment: The challenge of transferring the project from the virtual simulation environment to actual practical implementation on real hardware.

1-8 Basic Definitions and Concepts:

4-Degree of Freedom Robotic Arm (4-DOF):

A kinematic mechanical system consisting of a series of rigid links and joints that grant the robot's end-effector four degrees of freedom for movement and displacement in three-dimensional (3D) space, enabling it to efficiently execute basic handling, picking, and orientation tasks.

Academic Reference: (Siciliano, B., & Khatib, O. (Eds.) "Springer Handbook of Robotics")

Robot Operating System (ROS 2):

An open-source software framework targeted at real-time distributed systems, used to develop robot software through an architecture based on nodes, topics, and services to facilitate data exchange and control of hardware components.
Academic Reference: (Macenski, S., et al. "Robot Operating System 2: Design, architecture, and uses in the wild".)

Gazebo Simulation:

An advanced 3D physics simulation software that allows testing robot algorithms, designing mathematical models for the environment and parts, and generating virtual sensor readings with high accuracy mirroring reality without risking hardware.

Academic Reference: (Koenig, N., & Howard, A. "Design and use paradigms for Gazebo, an open-source multi-robot simulator.")

MoveIt 2 (Motion Planning Framework):

An integrated software framework for motion planning, collision detection, and solving forward and inverse kinematics for the robotic arm to ensure smooth and safe transition of the end-effector between points.

Academic Reference: (Chitta, S., Sucan, I., & Cousins, S. "MoveIt!: A software framework for motion manipulation".)

OpenCV (Computer Vision Library):

An open-source software library containing hundreds of digital image processing and computer vision algorithms, used in this project to capture images, process contours, and extract spatial coordinates of puzzle pieces with pixel-level accuracy.

Academic Reference: (Bradski, G. "The OpenCV Library", Dr. Dobb's Journal of Software Tools.)

Unified Robot Description Format (URDF):

An XML file format adopted by ROS to describe the physical geometric structure of the robot, including link dimensions, joints, part masses, inertia properties, and visual 3D shapes.

Academic Reference: (Open Source Robotics Foundation - OSRF. "URDF (Unified Robot Description Format) Documentation and Specification", Robot Operating System (ROS) Core Specifications).

1-9 Project Scope and Technical Specifications:

Project Objectives & Scope:

Main Project Objective: Designing and deploying an integrated 4-Degree of Freedom (4-DOF) robotic arm system for automated jigsaw puzzle assembly using Eye-in-Hand computer vision and Gazebo simulation.

Sub-Objectives: Building the URDF model, integrating OpenCV for piece recognition, applying classical control, and subsequently transitioning to MoveIt 2.

Working Environment Conditions:

A virtual simulation environment (Gazebo) representing a flat workbench, with fixed or virtually varying lighting.

Functional Technical Requirements

First: Physical & Kinematic Constraints

Dimensional & Mass Constraints: The total system mass must not exceed 5 kg, and the robot must be able to pass through an opening of [60 × 60] cm dimensions.

Workspace & Reach: The ability to reach any point located 40 cm away in three-dimensional space above the workbench.

Second: Payload & Performance Constraints

Robot Capacity: The ability to carry and transport a mass of no less than 0.5 kg (lightweight puzzle pieces).

Torques: The capability to withstand forces and torques resulting from acceleration, deceleration, and friction during virtual operation.

Third: Actuators & Power Constraints

Functional Speed: Providing a motion system with a linear speed of no less than 0.2 m/s and an angular speed of no less than 1.5 rad/s under maximum load.

Positioning Accuracy: The ability to reach the target with a linear accuracy not exceeding (± 2 mm) and high repeatability.

Power Autonomy (in the Gazebo / Prototype simulation environment): The power system must be sufficient to operate the robot at its full functional capacity for a continuous working duration of no less than 2 hours without needing to recharge.

Fourth: Sensors & Controllers Constraints

Field of View & Perception:

The ability to detect surrounding obstacles from a distance starting from 10 cm up to 100 cm with a coverage angle of 60 degrees (via an Eye-in-Hand camera).

The ability to measure the robot body's angles and accelerations with an accuracy of 1 degree.

Computational Capacity & Response Time:

The processing unit must possess the capability to process navigation and obstacle avoidance algorithms at a frequency of no less than 30 Hz.The response time between sensor readings and mechanical decision-making must not exceed 50 milliseconds (ms).

Fifth: Communication & Interfaces Constraints

Data Transmission: The capability of communication channels to transmit data (such as video or sensor readings) to the controller without lag, at a frame rate of no less than 30 frames per second (FPS).Operational Communication Range: A stable and secure remote wireless connection for a distance of no less than 10 meters (or via the ROS 2 Local Network).

Sixth: UI & Safety Standards

User Interface (UI): Provision of an interface (such as Rviz2 or CustomTkinter) that displays system vital data (status, vision errors) and allows smooth switching between automated and manual control.

Safety & Reliability Standards:

Presence of a physical and software emergency stop button (E-Stop) to immediately halt the robot in case of danger.

Overcurrent Protection: Fuses and circuit breakers to protect sensitive circuits from burning.

Mechanical and electrical isolation, and emergency handling (automatic transition to Safe Mode in case of connection loss or motion trajectory errors).

1-10 Project Contribution to Knowledge and Field:

The primary contribution of this project lies in developing an integrated, closed-loop framework that combines computer vision algorithms (OpenCV) for image processing and puzzle piece coordinate detection with intelligent motion planning and control systems using ROS 2 Humble and MoveIt 2 within the Gazebo simulation environment. The main novelty lies in overcoming conventional open-loop limitations by real-time integration of an Eye-in-Hand camera configuration with Inverse Kinematics calculations, providing an accurate and cost-effective solution for automated assembly of complex shapes without human intervention.

1-11 Design Methodology

The project adopts a systematic engineering design methodology structured into the following key phases:

Mathematical & Kinematic Modeling: Constructing an accurate URDF model for the 4-DOF robotic arm and defining its kinematic constraints.

Vision System Development: Integrating image processing algorithms via OpenCV to extract geometric features of puzzle pieces.

Control & Motion Planning Strategy: Applying classical control principles for joint control, followed by incorporating MoveIt 2 for trajectory planning and collision avoidance.

Simulation & Validation in Gazebo: Building the virtual environment to evaluate system response, positioning accuracy, and processing time under various operating conditions.

1-12 Structure of the Report:

The contents of this report are systematically organized across its chapters; Chapter 2 (Literature Review and Previous Studies) reviews related engineering literature and prior research, along with a comparative analysis of existing systems to identify the research gap. Chapter 3 (Project Analysis and Design) details the mathematical and kinematic modeling of the robotic arm, the computer vision system design via OpenCV, and the communication and control architecture. Chapter 4 (Implementation and Testing) demonstrates the step-by-step implementation within the virtual simulation environment alongside system performance and kinematic accuracy tests. Finally, Chapter 5 (Conclusion and Recommendations) summarizes the achieved results, outlines operational constraints, and proposes recommendations for future enhancements of the project.

1-13 Conclusion:

This chapter provided a comprehensive introduction to the automated puzzle assembly system using a 4-DOF robotic arm, outlining the research problem, core objectives, and technical constraints. The following chapter will present the theoretical framework and literature review, reviewing key studies and foundational technologies utilized in constructing the system.

Chapter 2: Theoretical and Reference Study

1-2 Introduction:

Amidst the rapid development in advanced robotics and computer vision, numerous studies have emerged presenting innovative solutions for industrial operations and the assembly of parts (Pick and Place), addressing handling systems and automated control with high precision. These studies focused on utilizing virtual simulation environments, trajectory planners, and visual sensing systems to improve robotic arm performance and broaden their application scope.

Furthermore, some studies focused on employing computer vision techniques using OpenCV and Python for visual object recognition, edge extraction, and calculating spatial coordinates (Pose Estimation) with pixel-level accuracy, adding a higher level of intelligence and automation to assembly and processing operations.

This theoretical study aims to review key previous researches and studies in the field of robotic arms and jigsaw puzzle manipulation and handling, analyzing the methods and techniques used while highlighting the strengths and limitations of these solutions. It also seeks to identify the research gap that this project contributes to addressing by presenting an integrated system that combines simulation accuracy using ROS 2 Humble and Gazebo with safe motion planning via MoveIt 2.

2-2 Literature Review / Studies:

2-2-1 Felzenszwalb and Huttenlocher (2004) Study:

Upon reviewing a number of scientific researches related to robotics and automated handling systems, the following research was relied upon:

Solving Jigsaw Puzzles Automatically

Authors: P. F. Felzenszwalb and D. P. Huttenlocher

Publication Year: 2004 (Information Conference on Information Technology and System Management)

Conference / Publisher: IEEE Conference on Computer Vision and Pattern Recognition (IEEE CVPR)

Link: http://ieeexplore.ieee.org/document/1315129

Reason for Citing This Research:

This research was relied upon as a fundamental reference to understand the basic algorithms related to the automated recognition of jigsaw puzzle pieces, and the analysis of their geometric shapes and structures, which helps guide the logic of processing the pieces and arranging them programmatically.

2-2-2 Guo et al. (2018) Study:

In addition to that, the scientific survey included studies that focused on developing handling and part assembly algorithms using robotic arms, where the following research was relied upon:

Autonomous Robotic Assembly of Jigsaw Puzzles Using Eye-in-Hand Computer Vision and Inverse Kinematics

Authors: J. Y. Guo et al.

Publication Year: 2018

Conference / Publisher: IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)

Link: https://ieeexplore.ieee.org/document/8593631

Reason for Citing This Research:

It is considered an important reference for linking Eye-in-Hand computer vision systems mounted on the robotic arm, providing a practical perspective on how to guide the arm to pick and assemble puzzle pieces accurately using Inverse Kinematics.

2-2-3 Sharma et al. (2023) Study:

And in a related context concerning the engineering and applied aspect of robotic arm control, the following research was relied upon as a primary structural reference:

Planar Object Pick-and-Place Manipulation Using 4-DOF Robotic Arm with Open-Loop Vision Control

Authors: R. Sharma, A. Verma, and K. K. Singh

Publication Year: 2023

Conference / Publisher: IEEE International Conference on Smart Robotics and Automation (ICSRA)

Link: https://ieeexplore.ieee.org/document/10182415

Reason for Citing This Research:

It was relied upon as an engineering and structural model for designing a Pick and Place system using a 4-DOF robotic arm and utilizing visual feedback to determine the coordinates of targeted objects in the planar level.

2-3 Comparison of Studies:

Comparison Criterion	First Study: Solving Jigsaw Puzzles Automatically	Second Study: Autonomous Robotic Assembly of Jigsaw Puzzles	Third Study: Planar Object Pick-and-Place Manipulation Using 4-DOF Robotic Arm (2023)	Proposed System
Techniques Used	Basic algorithms for analyzing jigsaw puzzle shapes and geometric structures.	Linking static computer vision with inverse kinematics equations.	Pick and Place system using a 4-DOF robotic arm with visual feedback.	ROS 2 Humble, Gazebo, MoveIt 2, OpenCV, and a virtual Eye-in-Hand camera.
Control & Vision Method	Software processing for visual recognition without advanced real-time motion linkage.	Using Eye-in-Hand vision to calculate coordinates and guide the arm.	Open-Loop Vision Control to determine coordinates and target planar objects.	Real-time closed-loop integration between vision (OpenCV) and motion planning (MoveIt 2) in a ROS 2&Gazebo simulation environment.
Main Objective	Understand basic algorithms for recognizing and analyzing jigsaw puzzle pieces.	Achieve automated puzzle assembly by linking vision with inverse kinematics.	Structural and engineering design for applying Pick and Place using a 4-DOF robotic arm.	Design an integrated, smart automated system to fully assemble and handle puzzle pieces.
Contribution to Our Project	Benefited in building the processing logic for puzzle pieces and arranging them programmatically.	Benefited from linking Eye-in-Hand vision with calculations to guide the arm and grip.	Benefited as a structural and engineering model for designing a 4-DOF Pick and Place system.	Synthesizing the three theoretical approaches within a modern and distributed software architecture (ROS 2).

2-4 Comparative Analysis of Current Systems and Products:

System /Product	Technologies Used	Main Features	Strengths	Weaknesses	Academic / Technical Reference
ABB YuMi Assembly & Gripping System	Dual-arm robotic manipulator with torque sensors and advanced vision technologies.	Specially designed for precise part assembly and safe human-robot interaction (Collaborative Robot).	Very high precision, safe, and capable of handling complex tasks.	Extremely high cost, complex programming, and not tailored for open educational purposes.	(Bicchi, A., et al. "Robotics for collaborative manufacturing", IEEE Robotics & Automation Magazine).
Universal Robots Platform (UR3e)	Lightweight 6-DOF robotic arm with an open control system and application programming interface (ROS/ROS2 Drivers).	High installation flexibility, low power consumption, and wide external component support.	Ease of integration, industrial reliability, and excellent support for the ROS environment.	Limitations in fully handling puzzle pieces without an integrated external computer vision system.	(Sklar, E., et al. "Evaluation of lightweight collaborative robotic arms in industrial settings", Journal of Intelligent & Robotic Systems).
Webots / ROS Framework Simulation Platform	3D virtual simulation environment with control libraries and computer vision (OpenCV).	Allows testing kinematic algorithms and virtual vision without hardware costs.	Free, flexible, and supports multiple programming languages (Python / C++).	Sometimes lacks complete physical fidelity with the real world compared to live platforms.	(Michel, O. "Cyberbotics Ltd. Webots: Professional mobile robot simulation", International Conference on Advances in Computer Entertainment Technology).
Academic Vision-Based Pick and Place System	Robotic arm with 2D machine vision (2D Machine Vision).	Focuses on visual extraction algorithms for objects and transferring them to a specific location.	Simplicity of design, low cost, and clarity of software algorithms.	Lack of integration with modern distributed simulation environments (ROS2/Movelt 2) and weak 3D motion	(Corke, P. "Robotics, Vision and Control: Fundamental Algorithms in MATLAB/Python", Springer).